IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-IT-066**

1. Topic (12 words max)

Decision Support System for Sri Lankan Paddy Cultivation to Optimize Financial Efficiency by Predicting disease stages and Aligning Cultivation & Harvest According to Future Supply & Demand

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-066
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lankan paddy agriculture is vital for national food security and rural livelihoods, especially for small and medium-scale farmers. Farmers face low productivity, inefficient irrigation and fertilizer use, frequent disease outbreaks (e.g., rice blast, brown spot), and income instability due to experience-based, fragmented decisions in irrigation, fertilization, and disease management without integrated data support.

Advancements in IoT, ML, and image processing offer smart solutions, but most address issues in isolation irrigation systems use fixed soil moisture thresholds, crop recommendations rely on static data, and image-based disease detection classifies without severity or economic assessment. These silo approaches fail to deliver holistic support.

Most ML models use non-Sri Lankan datasets, ignoring local conditions like monsoon rainfall, soil types, region-specific varieties (e.g., BG 300), and traditional practices, reducing accuracy. Systems rarely incorporate market price fluctuations, disconnecting decisions from economic returns and worsening instability.

Recent events like Cyclone Ditwah (2025) caused massive paddy losses (over 100,000 ha damaged/flooded), highlighting irrigation vulnerabilities in variable climate. An integrated, localized system combining real-time IoT, ML, image analysis, and market insights is essential for Sri Lankan paddy.

References:

[1] Department of Agriculture – Sri Lanka, "Crop Recommendations and Fertilizer Guidelines," 2021.

[2] Y. Wang, U. S. R. Dhamodharan, N. Sarwar, F. A. Almalki, and D. Mohan, "A hybrid approach for rice crop disease detection in agricultural IoT system," *Discover Sustainability*, vol. 5, art. no. 99, 2024.

[3] I. Abdurrahman and A. M. Miyim, "Machine Learning Model for Predicting Rice Crop Yield: A Case Study in Hadejia and Auyo, Nigeria," *UMYU Scientifica*, vol. 4, no. 1, pp. 239–249, 2025

[4] D. Pathirana and W. Wijesinghe, "Challenges in adopting smart farming technologies in Sri Lanka," *Journal of Agricultural Sciences – Sri Lanka*, 2020. *[Online]. Available: https://jas.sljol.info/*

6. Brief description of Existing Research and Systems (300 words max)

Several existing studies and agricultural guidelines attempt to address key challenges in crop cultivation, but they mostly operate in isolation. The **Department of Agriculture in Sri Lanka** provides official *Crop Recommendations and Fertilizer Guidelines* that help farmers follow standardized nutrient management and cultivation practices. While these guidelines are valuable, they are largely static and do not adjust to real-time field conditions such as soil variability, changing weather, or crop growth stages. They also lack predictive and economic decision-support features.

Research on **rice disease detection using IoT and machine learning** demonstrates how sensor data and image analysis can assist in identifying diseases at early stages. These systems improve classification accuracy; however, many focus only on disease recognition rather than assessing disease severity, predicting progression, or linking detection results with corrective actions like irrigation or fertilization adjustments. Additionally, localization for Sri Lankan farming conditions is often limited.

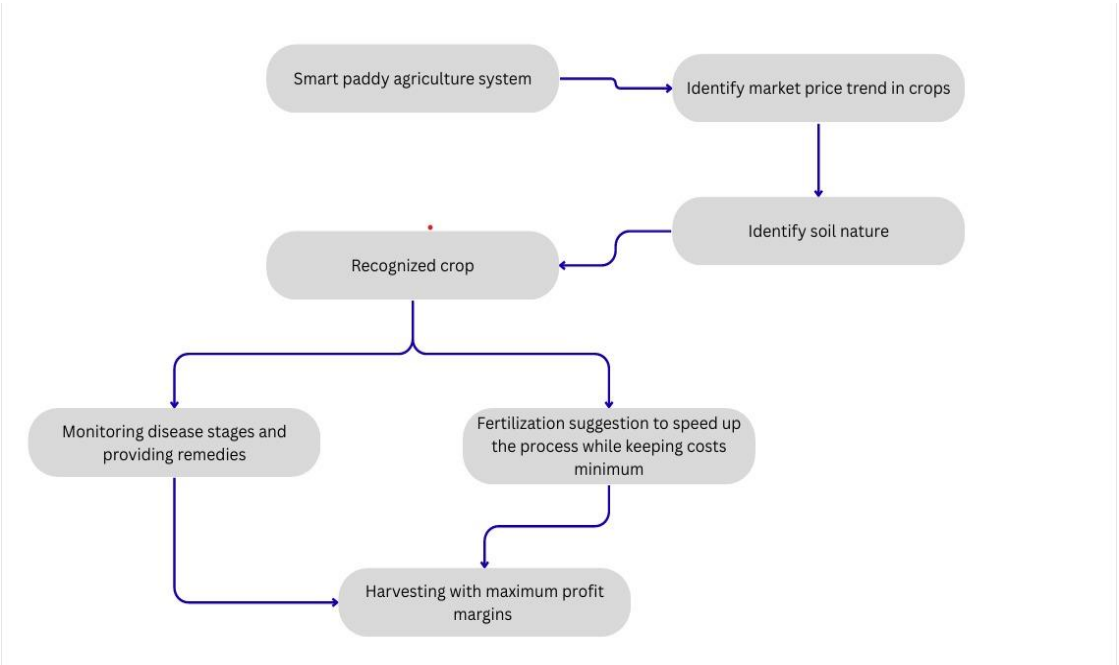
Similarly, **machine learning models for rice yield prediction** help estimate future production outcomes based on historical and environmental data . Although useful for forecasting, these models typically rely on region-specific datasets and are not tightly integrated with live IoT sensor inputs or farmer decision workflows.

Studies discussing **smart farming adoption challenges in Sri Lanka** highlight barriers such as cost, infrastructure, and knowledge gaps . While they explain why technologies struggle to scale, they do not propose integrated technical solutions.

Overall, existing approaches **address** fertilization, disease detection, yield prediction, and adoption **challenges separately**. There is still **no unified system** that connects **market forecasting, IoT based field assessment, growth-stage optimization, and disease severity analysis** specifically **for Sri Lankan** paddy farming.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

- The project is designed as a **Smart Agriculture Web Application** with a **cloud database**, integrating IoT sensing, crop health analysis, growth monitoring, and market intelligence into a unified platform.
- The **Market-Aware Crop Recommendation component** forecasts future supply and demand, helping farmers identify crops with higher potential profitability based on predicted market conditions.
- These market predictions are then **tallied with the IoT Field Monitoring component**, which evaluates soil and environmental parameters (pH, humidity, etc.) to determine whether the recommended crop is **agronomically suitable** for the field.
- Once cultivation begins, the **Growth Stage Monitoring component** continuously analyzes soil conditions and plant development to recommend **optimized fertilizer strategies**, ensuring efficient resource utilization and healthy crop growth.
- Simultaneously, the **Early Disease Detection and Severity Analysis component** monitors crop health, enabling early identification of diseases and timely preventive actions to **avoid yield reduction**.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The proposed Smart Agriculture system requires access to both domain-specific knowledge and reliable datasets to function effectively. For paddy cultivation in Sri Lanka, the **Department of Agriculture provides guidance on crop recommendations, fertilizer application, and cultivation practices**. These official guidelines form a foundational knowledge base that can be used to align the system's recommendations with local agronomic best practices.

In terms of **domain expertise**, collaboration with a lecturer, agricultural officers, local paddy farmers, and agronomy specialists is essential. Their practical experience provides context-specific knowledge that complements sensor data, such as optimal planting times, region-specific soil characteristics, and common disease patterns. Engaging with these experts ensures that system recommendations are realistic, actionable, and adapted to the Sri Lankan context.

Several datasets are available or can be generated to support the system. IoT sensors deployed in paddy fields will collect real-time soil and environmental data, including pH, moisture, NPK levels, temperature, humidity, and water levels**. Climate and weather data** can be obtained from the **Sri Lanka Meteorological Department**, providing rainfall patterns, temperature trends, and forecasts to assist in irrigation and growth-stage planning. **Disease image datasets** are required for early detection and severity analysis. These can be built through field collection in collaboration with local farmers and by **using publicly available rice disease datasets**, which will then be fine tuned for Sri Lankan disease variants.

Market and economic data are also necessary for the market-aware crop recommendation module. Historical supply and demand trends, current pricing, and crop profitability information can be sourced from **the Hector Kobbekaduwa Agrarian Research Institute** and **government agricultural databases**. This enables the system to provide economically informed recommendations rather than purely agronomic advice.

Overall, the combination of IoT sensor data, climate information, disease datasets, and market statistics, together with the knowledge and guidance of agronomy specialists and farmers, provides sufficient resources to develop a functional and reliable decision-support system. While some data may require field collection and preprocessing, the availability of both structured and unstructured data, along with local expertise, ensures that the system can make context aware, actionable, and economically optimized recommendations for paddy farmers in Sri Lanka.

9. Objectives and Novelty

Main Objective To design and implement an integrated Smart Agriculture Decision Support System for Sri Lankan paddy cultivation that addresses the limitations of existing solutions. Current approaches rely on static fertilization guidelines, isolated IoT monitoring, disease detection without severity assessment, and disconnected yield or market predictions. This system unifies real time IoT-based field monitoring, growth-stage adaptive resource management, early disease detection with severity analysis, and market-aware crop recommendation to enable farmers to make data-driven decisions that optimize both productivity and profitability.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22178718	To develop a context-aware IoT-based field monitoring module that continuously captures and interprets real-time soil, environmental, and paddy field conditions based on crop type, growth stage, and seasonal patterns relevant to Sri Lankan agriculture.	• Deploy sensors (moisture, NPK, pH, temp, humidity, rainfall, light, wetness, water). • Design real-time acquisition/transmissi on. - Preprocess/normalize data. • Adaptive interpretation logic. • Dashboard integration.	Unlike conventional IoT-based agricultural monitoring systems that rely on fixed environmental thresholds, this module introduces a **context-aware and growth-stage adaptive sensing framework** specifically designed for Sri Lankan paddy cultivation. The system dynamically interprets soil and environmental parameters (moisture, pH, NPK, humidity, water level) by incorporating **crop growth stage, seasonal variations, and localized agronomic conditions**. This transforms raw sensor readings into actionable insights rather than simple monitoring dashboards. Additionally, the module integrates

			directly with market prediction and decision-support components, enabling **cross-module linking**, which is rarely present in existing Sri Lankan IoT irrigation or monitoring solutions.
IT21811104	Intelligent crop recommendation combining IoT, climate, market prices for economic viability.	• Analyze sensor/climate data. • Preprocess market datasets. • Feature engineering. • Train/evaluate ML models. • Recommendation engine ranking.	Existing crop recommendation systems primarily consider soil characteristics and climatic suitability. This module introduces novelty by integrating **market supply & demand forecasting with real-time IoT depicted field conditions**, enabling economically optimized agronomic decisions. The recommendation engine evaluates not only what crop can grow, but **what crop should be grown for maximum profitability and reduced market risk**. The fusion of predictive market intelligence with environmental suitability modeling represents a shift from traditional productivity-centric systems to **agro-economic decision support**, particularly within Sri Lankan paddy farming.

IT22261632	Trained disease analysis detecting/estimating severity with environmental integration for recommendations. This will interpret how much the loss will be in harvesting if failure to solve within a given timeline.	• Collect/label leaf images. • Train DL models. • Integrate humidity/wetness. • Severity classification/progression. • Generate management recommendations.	While there are **AI-based disease detection systems** focus solely on classification, this component advances the state of practice by linking **disease severity estimation and progression awareness** using both image analysis and environmental factors. The system moves beyond "disease present/absent" outputs to provide **risk-informed management recommendations (Loss occur if not managed now)**. Furthermore, localization for **Sri Lankan paddy diseases and climatic conditions** enhances practical reliability compared to generic datasets and image only models.
IT22266446	Growth stage-aware irrigation/fertilization optimizing resources via time-series and imagery.	• Analyse time-series data. • Predict stages from images/temporal data. • Model stage-specific needs. • Generate schedules. • Dashboard integration.	Traditional irrigation and fertilization strategies apply generalized schedules or static rules. This module introduces a **predictive, growth stage aware optimization mechanism** driven by **time-series** sensor data and crop development indicators. Resource recommendations adapt dynamically to plant stage, soil conditions, and environmental variability. This replaces fixed-rule farming with **precision, stage-sensitive decision-making**, improving input efficiency and sustainability in paddy cultivation.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				